

HOMEWORK 2

INTRODUCTION TO COMPLEX ANALYSIS

(MATH 4230-40), SPRING 2022

DUE: 02/04/2022

SECTION 1.3 - SOME ELEMENTARY FUNCTIONS

1. Find all the values of:

(a) $(-i)^i$

$$\begin{aligned}
 (-i)^i &= e^{i \log(-i)} \\
 &= e^{i(\log(|-i|) + i \arg(-i))} \\
 &= e^{i(\log(1) + (-\frac{\pi}{2} + 2\pi n)i)}, \quad n \in \mathbb{Z} \\
 &= e^{(-\frac{\pi}{2} + 2\pi n)i^2}, \quad n \in \mathbb{Z} \\
 &= e^{\frac{\pi}{2} - 2\pi n}, \quad n \in \mathbb{Z}
 \end{aligned}$$

(b) $(1+i)^{1+i}$

$$\begin{aligned}
 (1+i)^{1+i} &= e^{(1+i)\log(1+i)} \\
 &= e^{(1+i)(\log(|1+i|) + i \arg(1+i))} \\
 &= e^{(1+i)(\log(\sqrt{2}) + (\frac{\pi}{4} + 2\pi n)i)}, \quad n \in \mathbb{Z} \\
 &= e^{[\log(\sqrt{2}) - \frac{\pi}{4} - 2\pi n] + [\frac{\pi}{4} + 2\pi n + \log(\sqrt{2})]i}, \quad n \in \mathbb{Z} \\
 &= e^{\log(\sqrt{2}) - \frac{\pi}{4} - 2\pi n} \left(\cos\left(\frac{\pi}{4} + 2\pi n + \log(\sqrt{2})\right) + i \sin\left(\frac{\pi}{4} + 2\pi n + \log(\sqrt{2})\right) \right), \quad n \in \mathbb{Z} \\
 &= \sqrt{2} e^{-\frac{\pi}{4} - 2\pi n} \left(\cos\left(\frac{\pi}{4} + 2\pi n + \log(\sqrt{2})\right) + i \sin\left(\frac{\pi}{4} + 2\pi n + \log(\sqrt{2})\right) \right), \quad n \in \mathbb{Z} \\
 &= \sqrt{2} e^{-\frac{\pi}{4} - 2\pi n} \left(\cos\left(\frac{\pi}{4} + \log(\sqrt{2})\right) + i \sin\left(\frac{\pi}{4} + \log(\sqrt{2})\right) \right), \quad n \in \mathbb{Z}
 \end{aligned}$$

2. Is it true that $|a^b| = |a|^{|b|}$ for all $a, b \in \mathbb{C}$? Why or why not?

No, it is not true that $|a^b| = |a|^{|b|}$ for all $a, b \in \mathbb{C}$. To show this, it is sufficient to provide a counterexample. One such counterexample is given by taking $a = i$ and $b = i$, so that $|a|^{|b|} = 1^1 = 1$, while (taking the principal branch of the complex logarithm)

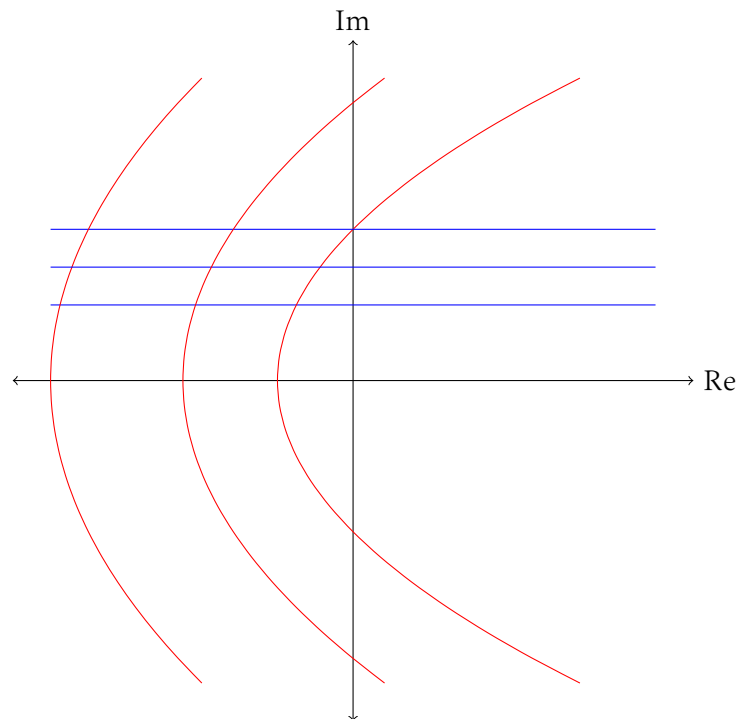
$$\begin{aligned}
 |a^b| &= |i^i| = |e^{i \log(i)}| \\
 &= |e^{i(\log(|i|) + i \arg(i))}| \\
 &= |e^{i(\log(1) + i \frac{\pi}{2})}| \\
 &= |e^{-\frac{\pi}{2}}| \approx 0.2078796
 \end{aligned}$$

3. Show that under the map $z \mapsto z^2$, lines parallel to the real axis are mapped to parabolas.

The equation of a line parallel to the x -axis is given by $y = c$ for some constant $c \in \mathbb{R}$. Writing $z \in \mathbb{C}$ as $z = x + iy$ so that $\operatorname{Re} z = x$ and $\operatorname{Im} z = y$, this equation then becomes $\operatorname{Im}(z) = c$. Hence, each line parallel to the real axis consists of points $z = x + ci$ for some constant $c \in \mathbb{R}$. Correspondingly, note that the map $z \mapsto z^2$ sends $z = x + ci$ to $z^2 = (x + ci)^2 = x^2 - c^2 + 2xci$. That is, $\operatorname{Re}(z^2) = x^2 - c^2$ and $\operatorname{Im}(z^2) = 2xc$. Hence,

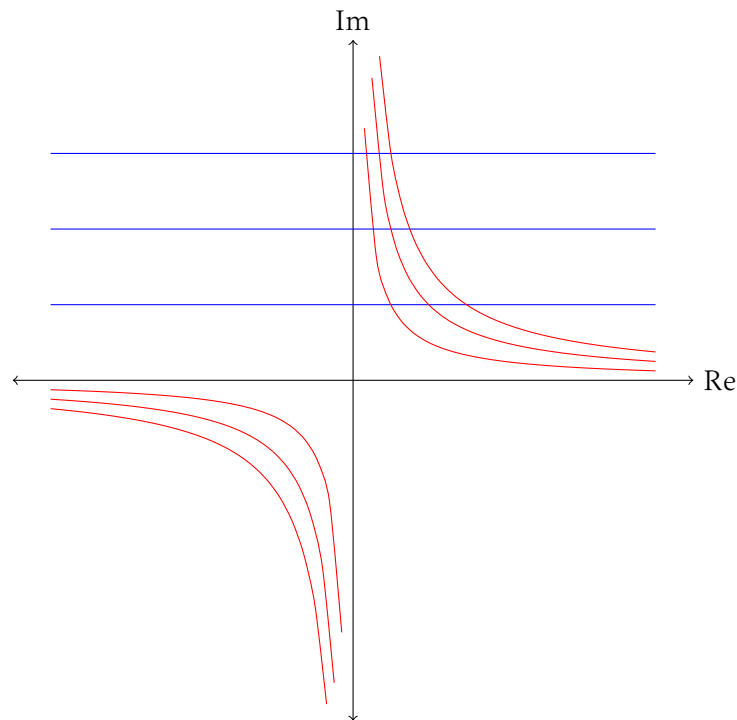
$$\operatorname{Re}(z^2) = \left(\frac{\operatorname{Im}(z^2)}{2c} \right)^2 - c^2.$$

This is a polynomial of degree two in $\operatorname{Im}(z^2)$. Graphically, this transformation sends the blue lines to the red parabolas below:



4. Show that under a branch of the map $z \mapsto \sqrt{z}$, lines parallel to the real axis are mapped to hyperbolas.

Using the answer from Problem 3, let us write $z^2 = u + vi$ so that $z \mapsto z^2$ sends pairs (x, y) to (u, v) . Since the squaring function is surjective by Proposition 1.1.3, every $z \in \mathbb{C}$ can be written in this form for some $(u, v) \in \mathbb{R}^2$. Therefore, a line parallel to the real axis consists of points $z = u + ci$ for some constant $c \in \mathbb{R}$; i.e. $v = c$. Since $v = 2xy$, this says that $c = 2xy$. Solving, this produces $y = \left(\frac{c}{2}\right) \frac{1}{x}$, which is a hyperbola. Visually, the squaring transformation sends the blue lines to the red hyperbolas below:



SECTION 1.4 - CONTINUOUS FUNCTIONS

5. Prove that the complement of any finite set of points in $\mathbb{C}$ is an open set.

Let $S \subseteq \mathbb{C}$ be a finite set of points. We desire to show that S^C is open.

To begin, we show that this holds in the case where $|S| = 1$ (e.g., when $S = \{s_0\}$). By Proposition 1.4.8, S is closed if and only if whenever $z_1, z_2, z_3, \dots$ is a sequence of points in S such that $\lim_{n \rightarrow \infty} z_n = z$ exists, then $z \in S$. Since the only sequence of points we can form in $S = \{s_0\}$ is the sequence given by $z_n = s_0$, it is certainly true that $z_n \rightarrow s_0 \in S$. Therefore, S is closed. As a closed subset of $\mathbb{C}$ is, by definition, the complement of an open set, it follows that $S^C = \mathbb{C} \setminus \{s_0\}$ is open.

Now, let us suppose that $|S| = n$; i.e., so that $S = \{s_0, s_1, \dots, s_n\}$. Notice that

$$\begin{aligned} S^C &= \mathbb{C} \setminus S \\ &= \mathbb{C} \setminus \{s_0, s_1, \dots, s_n\} \\ &= (\mathbb{C} \setminus \{s_0\}) \cap (\mathbb{C} \setminus \{s_1\}) \cap \dots \cap (\mathbb{C} \setminus \{s_n\}) \\ &= \{s_0\}^C \cap \{s_1\}^C \cap \dots \cap \{s_n\}^C. \end{aligned}$$

By Proposition 1.4.1, the intersection of any finite collection of open subsets of $\mathbb{C}$ is open. Since we have already shown that the complement of a set with one element is open, it then follows that S^C is open as well.

6. For what $z \in \mathbb{C}$ does the sequence (z_n) given by $z_n = nr^n$ converge?

Write $z = r(\cos \theta + i \sin \theta)$ and note that $z_n = nr^n(\cos n\theta + i \sin n\theta)$. If this sequence converges to some $z \in \mathbb{C}$, then it certainly cannot be the case that $\lim_{n \rightarrow \infty} |z_n| = \infty$. So, note that $|z_n| = nr^n$, which immediately yields that $\lim_{n \rightarrow \infty} nr^n = \infty$ whenever $r \geq 1$. Hence, it must be the case that $r < 1$ for any value z such that the sequence (z_n) converges.

Now, suppose that $z = r(\cos \theta + i \sin \theta)$ satisfies $r < 1$. We will show that $z_n \rightarrow 0$. To do so, notice that $|z_n| = nr^n = \frac{n}{(1/r)^n}$. As $\lim_{n \rightarrow \infty} n = \lim_{n \rightarrow \infty} (1/r)^n = \infty$, L'Hopital's rule tells us that

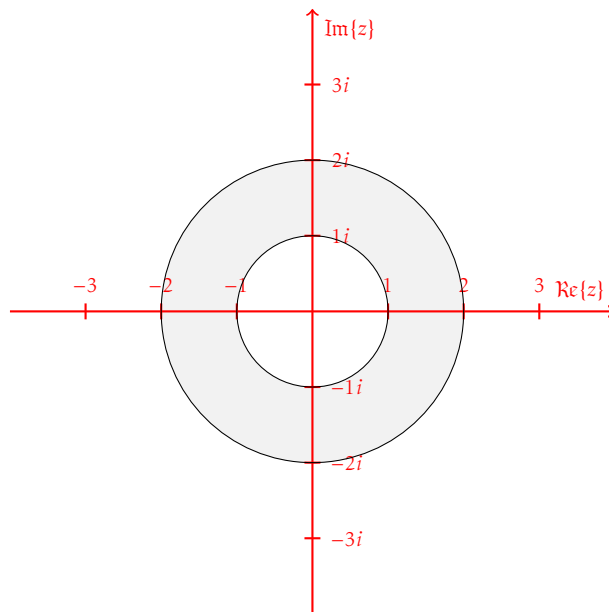
$$\begin{aligned} \lim_{n \rightarrow \infty} |z_n| &= \lim_{n \rightarrow \infty} nr^n \\ &= \lim_{n \rightarrow \infty} \frac{\frac{d}{dn}(n)}{\frac{d}{dn}((1/r)^n)} \\ &= \lim_{n \rightarrow \infty} \frac{1}{(1/r)^n \log(1/r)} \\ &= 0. \end{aligned}$$

Therefore, as $z = 0$ is the unique solution in $\mathbb{C}$ to $|z| = 0$, we may conclude that (z_n) diverges when $|z| \geq 1$ and $z_n \rightarrow 0$ whenever $|z| < 1$.

7. For each of the following sets, state whether or not it is connected and whether or not it is compact:

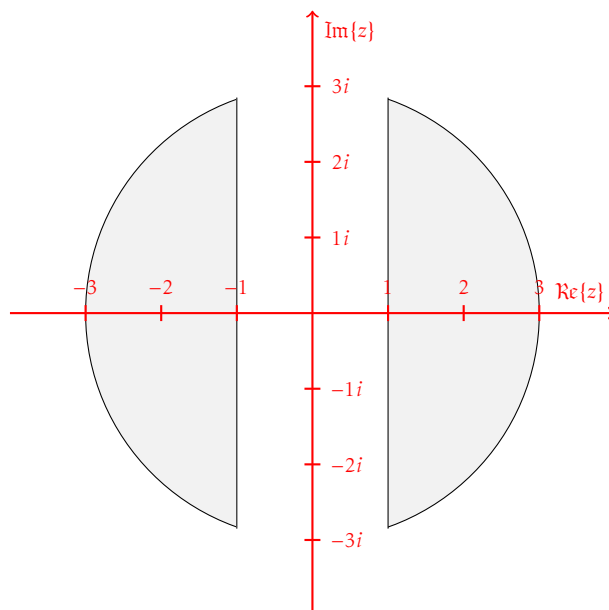
(a) $\{z \in \mathbb{C} : 1 \leq |z| \leq 2\}$

The set is connected and compact.



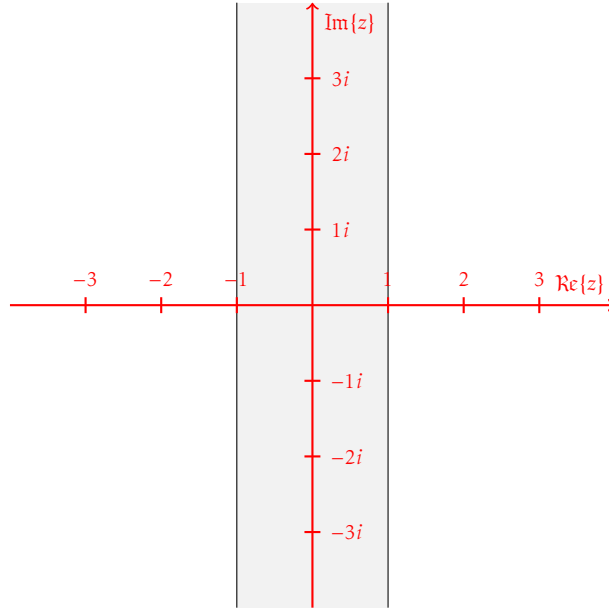
(b) $\{z \in \mathbb{C} : |z| \leq 3, |\text{Re} z| \geq 1\}$

The set is not connected, but is compact.



(c) $\{z \in \mathbb{C} : |\operatorname{Re} z| \leq 1\}$

The set is connected, but is not compact.



8. Use the fact that a function is continuous if and only if the preimage of every open set is open to show that a composition of two continuous functions is continuous.

Let (X, τ) , (Y, ν) , and (Z, η) be topological spaces and let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be continuous functions. Writing $g \circ f$ to denote the function $(g \circ f)(x) = g(f(x))$ We seek to show that $g \circ f : X \rightarrow Z$ is a continuous function. Using the fact that a function is continuous if and only if the preimage of every open set is open, this is equivalent to showing that if $\mathcal{O} \in \eta$ (that is, if $\mathcal{O}$ is an open set in Z), then $(g \circ f)^{-1}(\mathcal{O}) \in \tau$ (that is, the preimage of $\mathcal{O}$ is an open set in X). Notice that

$$\begin{aligned} (g \circ f)^{-1}(\mathcal{O}) &= \{x \in X : g(f(x)) \in \mathcal{O}\} \\ &= \{x \in X : f(x) \in g^{-1}(\mathcal{O})\} \\ &= f^{-1}(g^{-1}(\mathcal{O})). \end{aligned}$$

Since g is a continuous function and $\mathcal{O} \in \eta$, it follows that $g^{-1}(\mathcal{O}) \in \nu$. Since f is a continuous function and $g^{-1}(\mathcal{O}) \in \nu$, it follows that $f^{-1}(g^{-1}(\mathcal{O})) \in \tau$. Therefore, if $\mathcal{O} \in \eta$, then $(g \circ f)^{-1}(\mathcal{O}) \in \tau$, and we have shown that the the composition of two continuous functions is continuous.